GENERAL SPECIFICATION G3.24

Protective Coating for Concrete Surfaces

CONTENTS

G3.24 PROTECTIVE COATING FOR CONCRETE SURFACES

G3.24.1 Scope

(a) This section of the specification covers surface preparation and protective coating for concrete surfaces (excluding cement mortar lining and factory applied bitumen enamel wrapping).

(b) Details of the locations of the internal and external faces of buildings to which protective coatings are to be applied to be submitted to S.O. for approval.

(c) Where not otherwise required by the specification, or by the coating Manufacturers recommendations, the preparation and protection of surfaces shall be carried out in accordance with the recommendations of BS 6150.

G3.24.2 Reference Standards

(a) Unless otherwise specified and approved by the S.O. preparation and protective coatings shall comply with the relevant Singapore standards and the Reference standards listed below where applicable.

Standard	Subject
BS EN ISO 4618	Paints and varnishes. Terms and definitions
BS 3416	Bitumen-based coatings for cold application, suitable for use in contact with potable water
BS 6150	Painting of buildings. Code of practice
BS EN 1504 part 1 to 10	Mineral Based Coating for Acid Protection of concrete surfaces

G3.24.3 Submissions by the Contractor

(a) Submissions that the contractor is required to make in relation to protective coatings is covered under the specification for painting.

G3.24.4 Materials

(a) The general specification for materials is given in General Specifications G8.

(i) Protective coatings for concrete, cement Mortar Rendering and Plaster:

1) PVA Copolymer Emulsion Paint

- Polyvinylacetate acrylic copolymer emulsion paint shall be water based single pack non-flammable with a solids content by volume of 45 to 50%. Minimum dry film thickness shall be 50 microns.

2) High Performance Elastomeric Textured Paint

- Elastomeric textured paint system shall be styrene and acrylic based modified to be waterproof, elastomeric, water vapour permeable and carbon dioxide diffusion resistant.
- The primer shall be styrene acrylic based modified to be water vapour permeable and carbon dioxide diffusion resistant. Minimum dry film thickness of 25 microns.

- The undercoat shall be flexible acrylic water based non-plasticised elastomeric modified to be water vapour permeable. Minimum dry film thickness of 350 microns.
- The finish coat shall be flexible acrylic water based modified to be water vapour permeable, carbon dioxide diffusion resistant, chloride ion diffusion resistant, fungus and algae resistant and ultra violet degradation resistant. Minimum dry film thickness of 50 microns.
- The water vapour transmission equivalent air layer thickness tested in accordance with ASTM E96-80 shall be less than 4 m for the coating system.
- The carbon dioxide diffusion equivalent air layer thickness tested in accordance with ASTM E96-80 shall not be less than 50 m for the coating system.

3) Chemical Hardener and Dust Proof Compound

- Surface hardener for application to hardened concrete or granolithic screed shall be transparent water soluble crystalline material based on magnesium silicofluoride or sodium silicate.

4) Epoxy Coatings

- Epoxy protective coatings shall be of an approved high build polyamide cured epoxy, 2 or 3 pack, consisting of base and catalyst components completely resistant to the corrosion conditions to be encountered.
- The primer shall be a low viscosity two or three pack epoxy supplied by the manufacturer of the approved epoxy coating material. It shall have complete compatibility and inter-coat adhesion with the first coat of high build epoxy surface protective coating.
- The total dry film thickness of the protective coat shall have a minimum value of 750 microns.
- The epoxy resin coatings shall be so formulated that the epoxy resin content, together with its curing agent, shall be not less than 40% by weight of the solid binder.

5) Glass Flake Reinforced Coatings

- Glass flake reinforced coating systems shall comprise a two component vinylester peroxide hardened catalysed resin prime coat, a silica filled vinylester trowel applied base coat, a resin saturated glass fibre reinforcing mat (chopped strand mat) of 0.5 kg/m^2 weight and a two-component high solid glass flake reinforced vinylester resin surface coating as a finish, or equivalent system.

- The final system coating shall have a thickness not less than 2.5 mm and a surface hardness of 30 Barcol measured in accordance with BS 2782. The detailed system shall be submitted for the approval of the S.O..
- The glass reinforcement shall be of hardened glass flakes which shall, upon application, self-leaf and laminate to form a barrier of glass within the film. The finish coat shall contain hard alumina fillers in coatings applied to areas of high wear as directed by the S.O..

6) Cement Paints

- Cement paints shall comply with BS 4764; and
- Stopping for concrete or sand/cement plastering shall be of similar material to the background and shall have a similar surface finish.

7) Mineral Based Coating for Acid Protection of Concrete Surfaces

- The requirements described below are specified for establishing minimum quality standards. Products equal in quality to, or better than those specified, will be considered acceptable subject to approval by the S.O..
- The coating system and associated materials shall be supplied by a single manufacturer. If any product by a different manufacturer is proposed, submit all manufacturers' written confirmation that materials are compatible.
- The coating system shall comply with BS EN 1504.
- The coating system shall be two-component, mineral based for acid protection and shall have polymer silicates as binder.
- Specific products as recommended by the manufacturer shall be applied for various concrete surfaces listed as below:
  - Vertical surfaces outside of liquid zone;
  - Horizontal surfaces outside of liquid zone;
  - Vertical surfaces within liquid zone exposed constantly to wastewater; and
  - Horizontal surfaces within liquid zone exposed constantly to wastewater.
- The coating system shall have:
  - Very high chemical resistance against acids and solvents in the pH range of 0 to 8;
  - Very good adhesive on mineral based substrates (concrete, brickwork) and steel; and
  - High resistance against mechanical impact and abrasion, applicable by wet spraying technique onto vertical surfaces.

- The following are some of the minimum requirements of the spray applied coating system:
  - Layer thickness – 8 to 12 mm;
  - Tensile Strength at 23^0C and 50% relative humidity:
    - 4 N/mm^2 after 24 hours;
    - 7 N/mm^2 after 7 days; and
    - 12 N/mm^2 after 28 days.
  - Compressive Strength at 23^0C and 50% relative humidity:
    - 15 N/mm^2 after 24 hours;
    - 35 N/mm^2 after 7 days; and
    - 55 N/mm^2 after 28 days.
  - Pull-off strength at 23^0C and 50% relative humidity:
    - 1 N/mm^2 - lowest value; and
    - 1.5 N/mm^2 at 23^0C and 50% relative humidity.
  - Dynamic E modulus at 28 days:
    - 32,000 N/mm^2.
- Submittals
  - The Contractor shall submit the product data sheet including the manufacturer's specifications, installation instructions;
  - Samples of mineral based coating applied on concrete panels of 300 x 300 mm square; and
  - Manufacturer's Product History: Provide a list of not less than 5 projects where the system has been installed in similar used water conditions. List shall include project name, date of installation, employer, contact person and phone number.
- Warranty
  - Warranty stating that coating system installed will be free from defects for a period of not less than fifteen years from date of substantial completion of the Works. In the event of any defects found within the period stipulated, the Contractor shall, at the convenience of the Board, effect all repairs and replacements necessary to remedy defects all to the complete satisfaction of the S.O. at no additional cost to the Board.
- Product Storage and Handling
  - Products shall be delivered in their original, unopened packages, all clearly labelled with the manufacturer's name, brand name, and number and batch number of the material where appropriate, type and class as applicable, and the date of manufacture and expiration (if any);

- Products shall be stored as directed in a neat and safe manner. The storage area shall be shaded, protected from rain and surface water, ventilated and maintained at 8^0C to 20^0C. The product shall be located away from all sources of excess heat, sparks or open flame. Containers of liquid material shall not be left open at any time in the storage area;
- Materials not conforming to these requirements will be rejected by the S.O. and shall be removed from the site by the Contractor and replaced with approved materials, at no additional cost to the Board; and
- All safety precautions on product labels shall be observed. Containers shall not be welded, heated or drilled. All caps or bungs shall be replaced, and empty containers disposed from site.

(b) Miscellaneous

(i) Bituminous Coatings

1) Bituminous Coatings shall be formed of the appropriate grades of materials complying with BS 3416 or BS 6949 for cold application.

2) To avoid the possibility of the presence of carcinogenic polyaromatic hydrocarbons all bituminous paints and coatings must be manufactured from Petroleum or asphaltic bitumen and not from coal or for bitumen.

G3.24.5 Workmanship - General

(a) Preparation of Plaster, Brickwork and Concrete Surfaces

(i) Efflorescence present on the surfaces of plaster, brickwork and concrete shall be removed by scraping, brushing and wiping with a dry coarse cloth followed by a damp cloth. When efflorescence has been removed surfaces shall be left for at least three days and approved by the S.O. before surface protective coating. Surface protective coating shall be deferred where further salt deposits form on the surface during this period.

(ii) Plaster surfaces to be surface protective coated shall be cleaned down, filled and smoothed as necessary.

(iii) Brickwork, blockwork and concrete surfaces shall be cleaned of all contaminating matter before being surface protective coated. Subject to the approval of the S.O. large holes which would cause a break in the surface protective coating film shall be filled with mortar the surface being rubbed down to match the surrounding areas.

(b) Preparation of Concrete and Rendered Surfaces

(i) Concrete and rendered surfaces shall be thoroughly cured before the application of an epoxy filler primer or surface protective coating is begun.

(ii) All surfaces to be protected with a protective or decorative coating system shall be prepared as described below, maintaining the sequence of operations indicated.

(iii) Areas of concrete contaminated with machine oil or grease shall be cut out as necessary to remove all traces of such substances and shall be made good with an approved epoxy mortar.

(iv) Areas contaminated with release agent shall be scrubbed with suitable emulsion cleaners and any mould growth shall be treated with water-soluble fungicide. All surfaces so treated shall be rinsed off with water until they are clean of any mould growth and any other water soluble on the surfaces (e.g. salt).

(v) All concrete and rendered surfaces to be coated shall be lightly blast cleaned to remove the cement-rich surface layer, taking care not to roughen the surface unduly. Grit and detritus shall be removed by vacuum cleaning immediately prior to priming. All square edges and external corners are to be removed using suitable equipment. Coating grooves are to be formed in all internal corners. Before the application of the mineral bonding coat, the substrate must be pre-wetted carefully.

(vi) Blow-holes and honeycombed areas in the concrete which in the opinion of the S.O. are not capable of being satisfactorily levelled at the primer stage shall be filled with an epoxy mortar supplied or equivalent product recommended by the coating manufacturer for the purpose. Such mortar shall be knifed into the surface to level the area and to leave no excess.

(vii) Concrete surfaces which are to receive glass fibre/glass flake reinforced vinylester coatings shall be abraded by sand blasting and/or scarifying and shall have been allowed to cure for at least 28 days.

(viii) Coating shall not proceed until all drains, piping, conduit, vents, ducts and other projections through the substrate have been installed.

(ix) Sounding by Tapping: Testing of the entire surface by tapping in order to find all damages and cavities and the marking the areas concerned shall be carried out prior to the application of the coating.

(x) All loose and damaged concrete sections have to be chiselled out down to the sound core concrete.

(c) Installation procedures shall be in accordance with the manufacturer's standards.

G3.24.6 Application

(a) Coating shall be executed by an approved specialist applicator having a minimum of five years successful experience in the installation/application of the specified material. Only tradesmen experienced with the installation/application of the materials specified shall be employed.

(b) Certifications: Provide written statements of training and experience for individuals who will perform the installation.

(c) The Contractor must provide qualifications of his applicator who must be referred by name. The qualification certificate must confirm that the person concerned has adequate specialist and technical knowledge regarding the workmanlike execution of protection and repair measures on concrete surfaces, specialist and technical knowledge concerning the materials and tools to be used for the coating works.

(d) The Contractor's applicator shall submit the training certificate issued by the coating product manufacturer to ensure correct and skilled application with defined machinery and defined application conditions of the product.

(e) Environmental Conditions

  (i) Coating shall not be applied during inclement weather or when the air temperature is outside the range recommended by the manufacturer.

(f) Over the cleaned concrete or concrete blockwork surface the primer (bonding coat) shall be applied in accordance with the manufacturer's printed instructions. The coating shall be covered with protection board to protect it from damage during construction if found necessary. Areas around piping and protrusions shall be provided with an additional layer of coating system for a minimum of 300 mm in each direction.

(g) The impermeability of the surface protection shall be determined as per manufacturer's recommendations.

(h) Local ordinances and fire regulations shall be complied with in the installation of hazardous materials specified or required under this section. All necessary precautions shall be taken against fire and other hazards during delivery, storage and installation of flammable materials specified herein.

(i) Conform to all requirements limiting emissions of dust and volatile organic compounds.

END OF SECTION